

AWS

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "eks.amazonaws.com"
      },
      "Action": [
        "sts:AssumeRole",
        "sts:TagSession"
      ]
    }
  ]
}
```

```
aws iam create-role \
  --role-name AmazonEKSAutoClusterRole \
  --assume-role-policy-document file://trust-policy.json
```

```
aws iam get-role --role-name AmazonEKSAutoClusterRole --query "Role.Arn" --
output text
```

```
aws iam attach-role-policy \
  --role-name AmazonEKSAutoClusterRole \
  --policy-arn arn:aws:iam::aws:policy/AmazonEKSClusterPolicy
```

```
aws iam attach-role-policy \
  --role-name AmazonEKSAutoClusterRole \
  --policy-arn arn:aws:iam::aws:policy/AmazonEKSComputePolicy
```

```
aws iam attach-role-policy \
  --role-name AmazonEKSAutoClusterRole \
  --policy-arn arn:aws:iam::aws:policy/AmazonEKSBlockStoragePolicy
```

```
aws iam attach-role-policy \
  --role-name AmazonEKSAutoClusterRole \
  --policy-arn arn:aws:iam::aws:policy/AmazonEKSLoadBalancingPolicy
```

```
aws iam attach-role-policy \  
  --role-name AmazonEKSAutoClusterRole \  
  --policy-arn arn:aws:iam::aws:policy/AmazonEKSNetworkingPolicy
```

node-trust-policy.json:

```
{  
  "Version": "2012-10-17",  
  "Statement": [  
    {  
      "Effect": "Allow",  
      "Principal": {  
        "Service": "ec2.amazonaws.com"  
      },  
      "Action": "sts:AssumeRole"  
    }  
  ]  
}
```

```
aws iam create-role \  
  --role-name AmazonEKSAutoNodeRole \  
  --assume-role-policy-document file:///node-trust-policy.json
```

```
aws iam get-role --role-name AmazonEKSAutoNodeRole --query "Role.Arn" --  
output text
```

```
aws iam attach-role-policy \  
  --role-name AmazonEKSAutoNodeRole \  
  --policy-arn arn:aws:iam::aws:policy/AmazonEKSWorkerNodeMinimalPolicy
```

```
aws iam attach-role-policy \  
  --role-name AmazonEKSAutoNodeRole \  
  --policy-arn arn:aws:iam::aws:policy/AmazonEC2ContainerRegistryPullOnly
```

```
aws eks create-cluster --name demo-cluster \  
  --role-arn "arn:aws:iam::<account-id>:role/AmazonEKSAutoClusterRole"
```

```
--role-arn "arn:aws:iam::<account-id>:role/AmazonEKSAutoNodeRole"\n--resources-vpc-config subnetIds=subnet1,subnet2,subnet3
```

```
eksctl create cluster --name=<cluster-name> --enable-auto-mode
```

```
apiVersion: eksctl.io/v1alpha5\nkind: ClusterConfig\nmetadata:\n  name: demo-cluster\n  region: us-east-1\niam:\n  serviceRoleARN: arn:aws:iam::<account-id>:role/AmazonEKSAutoClusterRole\n\nautoModeConfig:\n  enabled: boolean\n  # To disable creation of nodePools, set it to the empty array ([]).\n  nodePools: []string\n  nodeRoleARN: arn:aws:iam::<account-id>:role/AmazonEKSAutoNodeRole
```

```
eksctl create cluster -f cluster.yaml
```

AKS

```
az aks create --resource-group demo-rg --name demo-cluster
```

GKE

```
gcloud container clusters create demo-cluster \n  --zone us-central1-a \n  --num-nodes 2 \n  --machine-type e2-micro \n  --enable-autoupgrade \n  --enable-autorepair
```

```
gcloud container clusters get-credentials demo-cluster --zone us-central1-a
```

```
gcloud container clusters delete demo-cluster --zone us-central1-a
```

```
gcloud container clusters create-auto demo-auto-cluster --location us-central1
```

IKS

```
ibmcloud ks cluster create vpc-gen2 --name demo-cluster --zone us-south-1 --flavor  
bx2.4x16 --workers 3 --vpc-id <vpc-id> --subnet-id <subnet-id>
```

```
ibmcloud ks cluster config --cluster demo-cluster
```

```
ibmcloud ks clusters
```

```
ibmcloud ks workers --cluster demo-cluster
```

```
ibmcloud ks worker-pool resize
```

```
ibmcloud ks cluster rm --cluster demo-cluster
```